

Enhanced Funnel Lifecycle & Measurement Overview

This document provides a comprehensive audit of the end-to-end conversion funnel, highlighting real user volumes, stage-by-stage drop-off rates, technical root causes in user session behavior, and critical data health and attribution vulnerabilities.

1. Executive Summary & Funnel Lifecycle Matrix

The matrix below details the performance metrics and measurement health across all five major stages of the user acquisition and conversion lifecycle.

Funnel Stage	Real User Volume	Conversion & Drop-off Rates	Session Behavior & Technical Root Cause	Analytics Status & Data Health
1. Acquisition (Ad Clicks)	7,486 clicks	Baseline	Healthy initial ad engagement (~14% CTR on reach).	Distorted: UTM parameters are stripped immediately downstream.
2. Course Evaluation (Preview)	~3,800 viewers	46.2% redirect loss	11-second average duration (vs. 4m+ for organic). Moodle guest security forces 4,084 mobile ad sessions from preview_courses.php to homepage /.	Broken: Stripped utm_* and fbclid cause GA4 to misclassify 4,179 sessions (74.5%) as 'Paid Other' or 'Direct'.
3. Lead Capture	3,452 registered	53.9% form	Multi-step mobile	Overstated:

Funnel Stage	Real User Volume	Conversion & Drop-off Rates	Session Behavior & Technical Root Cause	Analytics Status & Data Health
(Sign-Up)		drop-off	registration fatigue following a broken landing experience.	Meta pixel fires on "Register" button click, counting clicks instead of confirmed backend database writes.
4. Portal Activation (First Login)	269 active users	 92.2% never log in	3,183 accounts sit idle. No automated SMS, WhatsApp, or email onboarding nudges exist post-registration.	Blindspot: Tracking ceases after registration; active vs. idle users are unmonitored in ad channels.
5. Conversion (Payment)	25 paid buyers	9.3% conversion (of active users)	Successful M-PESA/Bank checkout for active users (KES 19,333.00 total).	Duplicated: Receipt page re-renders log 33 GA4 purchase events for only 4 buyers (KES 9,900), missing 21 real payers entirely.

2. Stage 2 Deep-Dive: Engagement & Attribution Leakage

The 11-Second Engagement Crash

Analysis reveals a severe anomaly between paid traffic and organic/direct sessions:

- **Paid Ad Traffic:** Averages just **8–11 seconds** in duration and **3.76–4.80 actions** per

session.

- **Direct / Organic Traffic:** Averages **4 minutes 1 second** in duration and executes **56.44 actions** per session.

The Redirect Root Cause

The primary catalyst for this drop-off is an infrastructure and security misconfiguration within Moodle:

1. Meta ad campaigns target `preview_courses.php` to show prospective users course previews.
2. Moodle's guest security configuration blocks unauthenticated mobile sessions attempting to view this page.
3. The system issues an automated **HTTP 302 redirect**, forcing mobile users away from the landing page onto the root homepage (`/`).
4. During the redirect chain, critical URL parameter data—including `utm_source`, `utm_medium`, and `fbclid`—is stripped completely.

Attribution Blindspot & Data Degradation

This architectural failure creates significant downstream attribution corruption:

- **4,084 sessions** are forcibly landed on the generic homepage instead of the intended course preview page.
- As a result, Google Analytics 4 (GA4) misclassifies **4,179 sessions (74.5%)** as generic 'Paid Other' or 'Direct'.
- This completely masks true advertising Return on Ad Spend (ROAS) while over-crediting 'Direct' traffic for returning user visits.

3. Stage 3–5 Technical & Analytics Vulnerabilities

Stage 3: Lead Capture Overstatement

The Meta Pixel is currently configured to trigger on the "Register" button click event rather than on verified backend database registration writes. This over-reports registration conversions and misleads ad optimization algorithms.

Stage 4: Post-Registration Activation Failure

A critical drop of **92.2%** occurs immediately after registration, leaving **3,183 accounts idle**. The

absence of automated onboarding communication (SMS, WhatsApp, or email) creates an operational blackout, exacerbated by zero user-level tracking beyond the registration step.

Stage 5: Conversion & Revenue Discrepancy

While 25 real users successfully completed M-PESA/Bank payments totaling KES 19,333.00, receipt page re-renders resulted in 33 duplicate GA4 purchase events logged for only 4 users (KES 9,900), leaving 21 actual paying users unrecorded in telemetry.